

# William S. Parker

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## SUMMARY

**Experimental physicist and R&D engineer** specializing in laser optics, signal processing, and multimodal imaging. I take a “full-stack” approach to R&D, with a proven ability to drive the end-to-end development of novel technologies. This ranges from high-level project design and pathfinding research, to hands-on mechanical, optical, and electronics prototyping, industrial control software development, to packaging lab-scale setups into robust, field-hardened systems.

## EDUCATION

### Ph.D., Physics

Awarded: *OMQ Director's Fellowship, Graduate First Year Fellowship*

University of Oregon

Dec. 2024

### B.S., Double Major in Physics and Mathematics

Awarded: *Outstanding Senior in Physics, Magna Cum Laude*

Chapman University

May 2018

## EXPERIENCE

### Jr. Research Physicist

May 2025 – present

*AstroForge, Inc., Seal Beach, CA*

- Designed and built optical trains in lab and field test settings, integrating high-energy (>200 mJ) pulsed lasers, high-power (4 kW) fiber lasers, motion control / fast steering, beam conditioning optics, and metrology hardware.
- Hardened optical setups for deployment in a real mine, protecting against dust, vibration, and thermal cycling.
- Designed and built internal controls software, industrial control electronics, and microcontroller I/O circuit boards to run precisely coordinated, flexible experiments with data acquisition pipelines.
- Rapidly prototyped optical setups and mechanical systems using 3D printing, laser-cut metal, custom electronics, and commercial optics.
- **High degree of independent ownership over:** high-level R&D planning, low-level experiment design, hiring (grew the team from just me to 9 people), lab / hardware bring-up, system architectures, instrumentation selection.

### PhD Candidate

Mar. 2020 – Dec. 2024

*University of Oregon, Eugene, OR*

- Developed holographic techniques for atomic-resolution magnetic imaging at Lawrence Berkeley National Lab.
- Experimentally determined the 3D structure of magnetic skyrmions by combining novel and conventional electron microscopy techniques with numerical modeling on a high-performance compute cluster.
- Developed algorithms to analyze, simulate, and visualize electron optics data, publishing packages to PyPI.
- Presented results at 6 international scientific conferences, initiating industry and academic collaborations.
- Wrote and published first-author works in Physical Review B and Microscopy & Microanalysis.

### Graduate Research Assistant

Jun. 2018 – Mar. 2020

*University of Oregon, Eugene, OR*

- Fabricated and tested diffractive electron optics elements using a focused ion beam microscope.
- Numerically modeled a free-space orbital angular momentum sorter using finite-element and Fourier optics.
- Presented results at the 2019 Microscopy & Microanalysis conference; earned \$1,000 MSA Student Scholar award.

### Undergraduate Researcher

Feb. 2017 – May 2018

*Chapman University, Orange, CA*

- Generated custom optical vortex modes using a digital micromirror device for Raman spectroscopy applications.

### Undergraduate Researcher (National Science Foundation REU, Joseph Eberly)

May. 2017 – Jun. 2018

*University of Rochester, Rochester, NY*

- Numerically investigated fundamental limitations and potential improvements to mode-locked ultrafast lasers.

## SKILLS

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- **Optical & Imaging Systems:** Laser optics (high-power, pulsed, ultrafast), beam shaping, Fourier optics, interferometry, electron microscopy (TEM/SEM), cryogenic imaging, spectroscopy (Raman, X-ray, IR).
- **Engineering & Prototyping:** Rapid prototyping (3D printing, laser-cut metal), mechanical design (CAD/Fusion), opto-mechanics, PCB design, analog/digital electronics, system hardening, system validation.
- **Controls & Integration:** Industrial automation, motion control, data acquisition, hardware/software integration, microcontroller I/O, ultra-high vacuum (UHV) systems.
- **Computation & Simulation:** Numerical physics modeling, finite-element analysis, charged particle optics, signal/image processing (OpenCV, FIJI, custom), machine learning.
- **Software:** Python, JS/Node/Svelte, Git, Mathematica, Blender, high-performance computing (HPC), Arduino, Fusion 360.
- **Leadership & Collaboration:** Cross-functional team leadership (scaled from 1 to 9 members), multidisciplinary R&D coordination, scientific communication (6+ international conferences), technical documentation.

## SOFTWARE DEVELOPMENT

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- [ltempy](#) / [GitHub](#) - Tools for the analysis, simulation, and presentation of Lorentz TEM data.
- [frctl](#) / [GitHub](#) - A single-page web app for exploring the Mandelbrot and Burning Ship fractals.

## SELECTED PUBLICATIONS

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- Hybrid Skyrmions in Magnetic Multilayer Thin Films are Half-Integer Hopfions. **W.S. Parker**, J. Reddinger, B.J. McMorran. *Phys. Rev. B*. **2024**. DOI: [10.1103/PhysRevB.110.224420](https://doi.org/10.1103/PhysRevB.110.224420).
- Real Space Imaging of Hybrid Skyrmion Textures in Magnetic Multilayer Thin Films. **W.S. Parker**, J. Reddinger, R. Moraski, E.E. Fullerton, S.A. Montoya, B.J. McMorran. *Phys. Rev. B*. **2025**. DOI: [10.1103/زلزپ-چدق](https://doi.org/10.1103/زلزپ-چدق)
- Combining Lorentz TEM and SEM with Polarization Analysis to Uncover Fractional Topological Spin Textures in Fe/Gd Multilayer Thin Films. **W.S. Parker**, S.A. Montoya, E.E. Fullerton, B.J. McMorran. *Microscopy and Microanalysis*. **2024**. DOI: [10.1093/mam/ozae044.507](https://doi.org/10.1093/mam/ozae044.507).
- Exact Design of Complex Amplitude Holograms for Producing Arbitrary Scalar Fields. C.W. Johnson, J.S. Pierce, R.C. Moraski, A.E. Turner, A.T. Greenberg, **W.S. Parker**, B.J. McMorran. *Opt. Express*. **2020**. DOI: [10.1364/OE.393224](https://doi.org/10.1364/OE.393224).

## AWARDS, GRANTS, SCHOLARSHIPS

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- Best Student Presentation – Finalist, Magnetism and Magnetic Materials. 2023
- Honorable Mention, National Science Foundation Graduate Research Fellowship. 2020
- Student Scholar Award, Microscopy and Microanalysis. 2019
- Graduate First Year Fellowship, University of Oregon. 2018-2019
- OMQ Director's Fellowship, University of Oregon. 2018
- Outstanding Senior in Physics, Chapman University. 2018
- Chancellor's Scholarship, Chapman University. 2014-2018
- Chapman Celebrates – Music, Chapman University. 2014-2018
- Golden Ears Award, Chapman University. 2016

## RELEVANT COURSEWORK

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- **Graduate (University of Oregon):** Quantum Information (PHYS 610), Quantum Optics and Laser Physics (PHYS 685), Solid State Physics (PHYS 672), Space Physics (ERTH 510), Scientific Computation (PHYS 510)
- **Undergraduate (Chapman University):** Quantum Information Science (PHYS 340)

## LINKS

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[wsparker.com](http://wsparker.com) | [LinkedIn](#) | [GitHub](#) | [Google Scholar](#) | [Curriculum Vitae](#)